I- Introduction

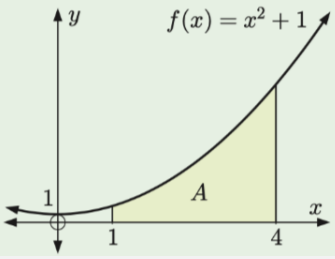
Investigation

The function $f(x) = x^2 + 1$ lies above the x-axis for all $x \in \mathbb{R}$.

Things to think about:

a How can we calculate the shaded area A, which is the area under the curve for $1 \leqslant x \leqslant 4$?

b What function has $x^2 + 1$ as its derivative?

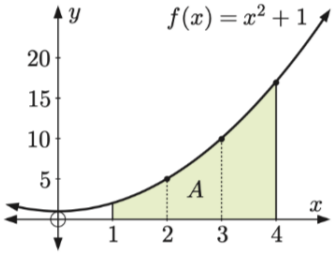


a) Ancient mathematicians way

Consider the function $f(x) = x^2 + 1$.

We wish to estimate the area A enclosed by $y = f(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 4$.

Suppose we divide the interval $1 \leqslant x \leqslant 4$ into three strips of width 1 unit as shown. We obtain three sub-intervals of equal width.



Upper rectangles :

The area of the upper rectangles,

$$A_U = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots \text{ units}^2$$

Lower rectangles :

The area of the lower rectangles,

$$A_L = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots \text{ units}^2$$

Now clearly $A_L < A < A_U$, so the area A lies between ……units^2 and ……units^2.

If the interval $1 \leqslant x \leqslant 4$ was divided into 6 subintervals, each of width $\frac{1}{2}$, then

$$\begin{aligned} A_U &= \tfrac{1}{2}f\left(1\tfrac{1}{2}\right) + \tfrac{1}{2}f(2) + \tfrac{1}{2}f\left(2\tfrac{1}{2}\right) + \tfrac{1}{2}f(3) + \tfrac{1}{2}f\left(3\tfrac{1}{2}\right) + \tfrac{1}{2}f(4) \\ &= \tfrac{1}{2}\left(\tfrac{13}{4} + 5 + \tfrac{29}{4} + 10 + \tfrac{53}{4} + 17\right) \\ &= 27.875 \text{ units}^2 \end{aligned}$$

and

$$\begin{aligned} A_L &= \tfrac{1}{2}f(1) + \tfrac{1}{2}f\left(1\tfrac{1}{2}\right) + \tfrac{1}{2}f(2) + \tfrac{1}{2}f\left(2\tfrac{1}{2}\right) + \tfrac{1}{2}f(3) + \tfrac{1}{2}f\left(3\tfrac{1}{2}\right) \\ &= \tfrac{1}{2}\left(2 + \tfrac{13}{4} + 5 + \tfrac{29}{4} + 10 + \tfrac{53}{4}\right) \\ &= 20.375 \text{ units}^2 \end{aligned}$$

From this refinement we conclude that the area A lies between 20.375 and 27.875 units^2.

As we create more subdivisions, the estimates A_L and A_U will become more and more accurate. In fact, as the subdivision width is reduced further and further, both A_L and A_U will converge to A.

Remark. *We can use technology to help calculate A_L and A_U for large values of n.*

The example of $y = x^2$ and the x-axis for $0 \leqslant x \leqslant 1$ is of historical interest. Archimedes (287 - 212 BC) found the exact area. In an article that contains 24 propositions, he developed the essential theory of what is now known as integral calculus.

EXERCISE 1

Consider the area between $y = x$ and the x-axis from $x = 0$ to $x = 1$.

a Divide the interval into 5 strips of equal width, then estimate the area using:
   i upper rectangles ii lower rectangles.

b Calculate the actual area and compare it with your answers in a.

EXERCISE 2

Consider the area between $y = \dfrac{1}{x}$ and the x-axis from $x = 2$ to $x = 4$. Divide the interval into 6 strips of equal width, then estimate the area using:

a upper rectangles b lower rectangles.

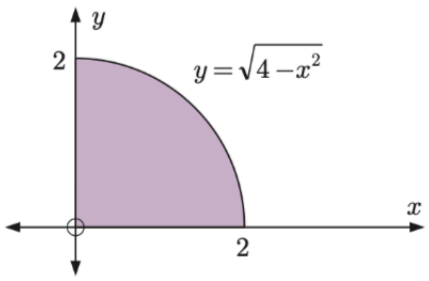
EXERCISE 3

Consider the quarter circle of centre $(0, 0)$ and radius 2 units illustrated.

Its area is $\frac{1}{4}$(full circle of radius 2)

$= \frac{1}{4} \times \pi \times 2^2$

$= \pi$ units^2



a Estimate the area using lower and upper rectangles for $n = 10, 50, 100, 200, 1000$, and $10\,000$. Hence, find rational bounds for π.

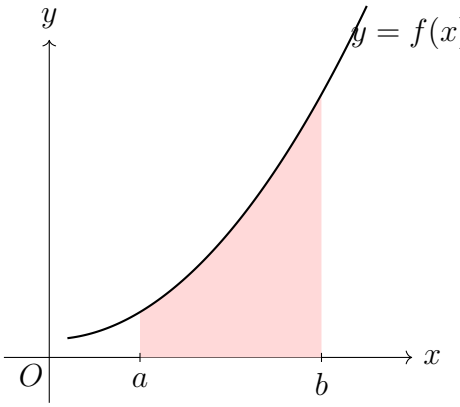
b Archimedes found the famous approximation $3\frac{10}{71} < \pi < 3\frac{1}{7}$.
For what value of n is your estimate for π better than that of Archimedes?

b) Some definitions

Definition

The value A is known as the "definite integral of $f(x)$ from a to b", written $A = \int_a^b f(x)\,dx$.

If $f(x) \geqslant 0$ for all $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx$ is equal to the shaded area.



Example

1. Sketch the graph of $y = x^4$ for $0 \leqslant x \leqslant 1$. Shade the area described by $\int_0^1 x^4\,dx$.
2. Use technology to calculate the lower and upper rectangle sums for n equal subdivisions where $n = 5, 10, 50, 100$, and 500.
3. Hence find $\int_0^1 x^4\,dx$ to 2 significant figures.

EXERCISE 4

a Sketch the graph of $y = \sqrt{x}$ for $0 \leqslant x \leqslant 1$.

Shade the area described by $\int_0^1 \sqrt{x}\, dx$.

b Find the lower and upper rectangle sums for $n = 5$, 10, 50, 100, and 500.

c Use the information in b to find $\int_0^1 \sqrt{x}\, dx$ to 2 significant figures.

Example

Use graphical evidence and known area facts to find :

1. $\int_0^2 (2x+1)\, dx$
2. $\int_0^1 \sqrt{1-x^2}\, dx$

EXERCISE 5

Use graphical evidence and known area facts to find:

a $\int_1^3 (1+4x)\, dx$ b $\int_{-1}^2 (2-x)\, dx$ c $\int_{-2}^2 \sqrt{4-x^2}\, dx$

II- Antidifferentiation

Examples of problems we need to solve include :

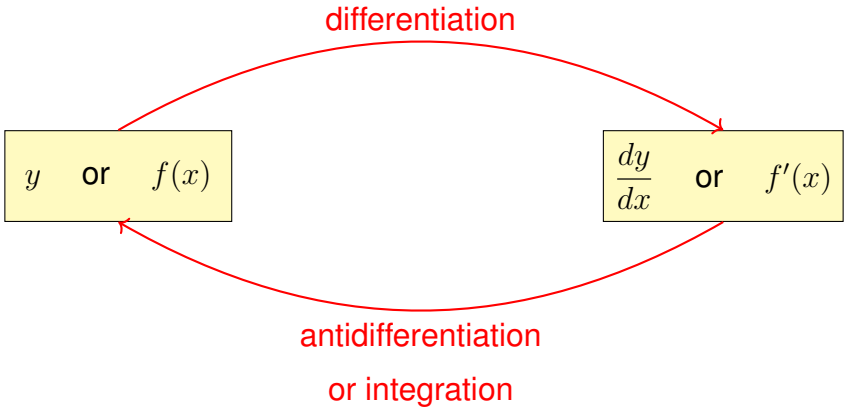
— The gradient function $f'(x)$ of a curve is $2x+3$ and the curve passes through the origin. What is the function $y = f(x)$?

— The rate of change in temperature is $\frac{dT}{dt} = 10e^{-t}$ °C per minute where $t \geqslant 0$. What is the temperature function given that initially the temperature was 11°C ?

a) Definition

Definition

The process of finding y from $\frac{dy}{dx}$, or $f(x)$ from $f'(x)$, is the reverse process of differentiation. We call it antidifferentiation.



Definition

If $F(x)$ is a function where $F'(x) = f(x)$ we say that :

— the derivative of $F(x)$ is $f(x)$ and

— the antiderivative of $f(x)$ is $F(x)$.

Example

Find the antiderivative of :

1. x^3
2. e^{2x}
3. $\frac{1}{\sqrt{x}}$

EXERCISE 6

a Find the antiderivative of:

i x **ii** x^2 **iii** x^5 **iv** x^{-2} **v** x^{-4} **vi** $x^{\frac{1}{3}}$ **vii** $x^{-\frac{1}{2}}$

b From your answers in **a**, predict a general rule for the antiderivative of x^n, for $n \neq -1$.

EXERCISE 7

a Find the antiderivative of:

i e^{2x} **ii** e^{5x} **iii** $e^{\frac{1}{2}x}$ **iv** $e^{0.01x}$ **v** $e^{\pi x}$ **vi** $e^{\frac{x}{3}}$

b From your answers in **a**, predict a general rule for the antiderivative of e^{kx} where k is a constant.

EXERCISE 8

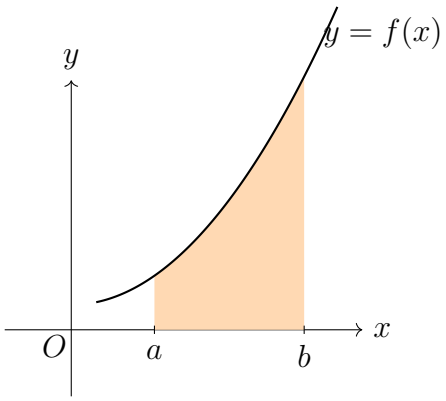
Find the antiderivative of:

a $6x^2 + 4x$ by first differentiating $x^3 + x^2$ **b** e^{3x+1} by first differentiating e^{3x+1}

c $\sqrt{x}$ by first differentiating $x\sqrt{x}$ **d** $(2x+1)^3$ by first differentiating $(2x+1)^4$

Definition

If $f(x)$ is a continuous positive function on an interval $a \leqslant x \leqslant b$ then the area under the curve between $x = a$ and $x = b$ is $\int_a^b f(x)\,dx$.



Definition

$$\int_a^t f(x)\,dx = F(t) - F(a) \quad \text{where} \quad F'(x) = f(x). \quad F(x) \text{ is the antiderivative of } f(x).$$

b) Fundamental theorem of calculus

Property

For a continuous function $f(x)$ with antiderivative $F(x)$,

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Property

- $\int_a^a f(x)\,dx = 0$
- $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$
- $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$
- $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
- $\int_a^b c\,dx = c(b-a)$ {c is a constant}
- $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$

Proof

$$\begin{aligned}\int_a^b f(x)\,dx + \int_b^c f(x)\,dx &= F(b) - F(a) + F(c) - F(b)\\ &= F(c) - F(a)\\ &= \int_a^c f(x)\,dx\end{aligned}$$

Example

Use the fundamental theorem of calculus to find the area between :

1. the x-axis and $y = x^2$ from $x = 0$ to $x = 1$
2. the x-axis and $y = \sqrt{x}$ from $x = 1$ to $x = 9$.

EXERCISE 9

Use the fundamental theorem of calculus to find the area between:

a the x-axis and $y = x^3$ from $x = 0$ to $x = 1$

b the x-axis and $y = x^2$ from $x = 1$ to $x = 2$

c the x-axis and $y = \sqrt{x}$ from $x = 0$ to $x = 1$.

Check your answers using technology.

EXERCISE 10

Use the fundamental theorem of calculus to show that:

a $\int_a^a f(x)\,dx = 0$ and explain the result graphically

b $\int_a^b c\,dx = c(b-a)$ where c is a constant

c $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$

d $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$ where c is a constant

e $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

EXERCISE 11

Use the fundamental theorem of calculus to find the area between the x-axis and:

a $y = x^3$ from $x = 1$ to $x = 2$

b $y = x^2 + 3x + 2$ from $x = 1$ to $x = 3$

c $y = \sqrt{x}$ from $x = 1$ to $x = 2$

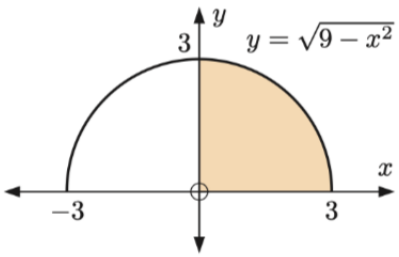
d $y = e^x$ from $x = 0$ to $x = 1.5$

e $y = \dfrac{1}{\sqrt{x}}$ from $x = 1$ to $x = 4$

f $y = x^3 + 2x^2 + 7x + 4$ from $x = 1$ to $x = 1.25$

EXERCISE 12

Use technology to find the area between the x-axis and $y = \sqrt{9-x^2}$ from $x = 0$ to $x = 3$.
Check your answer by direct calculation of the area.



III- Rules of integration

a) Indefinite integrals

Definition

This is process known as indefinite integration because it is not being applied to a particular interval.

If $F'(x) = f(x)$ then $\int f(x)\,dx = F(x) + c$.

Property

Any constant may be written in front of the integral sign.

$$\int k\,f(x)\,dx = k\int f(x)\,dx, \quad k \text{ is a constant}$$

Proof

Consider differentiating $kF(x)$ where $F'(x) = f(x)$.

$$\frac{d}{dx}\big(k\,F(x)\big) = k\,F'(x) = k\,f(x)$$

$$\therefore \quad \int k\,f(x)\,dx = k\,F(x)$$

$$= k\int f(x)\,dx$$

Property

The integral of a sum is the sum of the separate integrals. This rule enables us to integrate term by term.

$$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

Example

If $y = x^4 + 2x^3$, find $\frac{dy}{dx}$. Hence find $\int (2x^3 + 3x^2)\,dx$.

EXERCISE 13

If $y = x^3 + x^2$, find $\frac{dy}{dx}$. Hence find $\int (3x^2 + 2x)\,dx$.

EXERCISE 14

If $y = (2x+1)^4$ find $\frac{dy}{dx}$. Hence find $\int (2x+1)^3\,dx$.

Example

Suppose $y = \sqrt{5x-1}$.

1. Find $\frac{dy}{dx}$.
2. Hence find $\int \frac{1}{\sqrt{5x-1}}\,dx$.

Remark. *We can check that an integral is correct by differentiating the answer. It should give us the integrand, the function we originally integrated.*

EXERCISE 15

If $y = \frac{1}{\sqrt{x}}$, find $\frac{dy}{dx}$. Hence find $\int \frac{1}{x\sqrt{x}}\,dx$.

EXERCISE 16

By considering $\frac{d}{dx}(x^2-x)^3$, find $\int (2x-1)(x^2-x)^2\,dx$.

EXERCISE 17

Find $\frac{dy}{dx}$ if $y = \sqrt{1-4x}$. Hence find $\int \frac{1}{\sqrt{1-4x}}\,dx$.

b) Antiderivatives

Derivative table summary

Function	*Derivative*	*Name*
c, a constant	0	
$mx+c$, m and c are constants	m	
x^n	nx^{n-1}	power rule
$cu(x)$	$cu'(x)$	
$u(x)+v(x)$	$u'(x)+v'(x)$	addition rule
$u(x)v(x)$	$u'(x)v(x)+u(x)v'(x)$	product rule
$\dfrac{u(x)}{v(x)}$	$\dfrac{u'(x)v(x)-u(x)v'(x)}{[v(x)]^2}$	quotient rule
$y=f(u)$ where $u=u(x)$	$\dfrac{dy}{dx}=\dfrac{dy}{du}\dfrac{du}{dx}$	chain rule
e^x	e^x	
$e^{f(x)}$	$e^{f(x)}f'(x)$	
$\ln x$	$\dfrac{1}{x}$	
$\ln f(x)$	$\dfrac{f'(x)}{f(x)}$	
$[f(x)]^n$	$n[f(x)]^{n-1}f'(x)$	
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\dfrac{1}{\cos^2 x}$	

Property

$$\int k\,dx = kx + c$$

Proof

For k a constant, $\dfrac{d}{dx}(kx+c)=k$

Property

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}+c, \quad n\neq -1$$

Proof

If $n\neq -1$, $\dfrac{d}{dx}\left(\dfrac{x^{n+1}}{n+1}+c\right)=\dfrac{(n+1)x^n}{n+1}=x^n$

Property

$$\int e^x \, dx = e^x + c$$

Proof

$$\frac{d}{dx}(e^x + c) = e^x$$

Property

$$\int \frac{1}{x} \, dx = \ln|x| + c$$

Proof

If $x > 0$, $\quad \frac{d}{dx}(\ln x + c) = \frac{1}{x}$

If $x < 0$, $\quad \frac{d}{dx}(\ln(-x) + c) = \frac{-1}{-x} = \frac{1}{x}$

Property

$$\int \cos x \, dx = \sin x + c$$

Proof

$$\frac{d}{dx}(\sin x + c) = \cos x$$

Property

$$\int \sin x \, dx = -\cos x + c$$

Proof

$$\frac{d}{dx}(-\cos x + c) = \sin x$$

Remark. *c is an arbitrary constant called the constant of integration or integrating constant.*

Example

Find :

1. $\int (x^3 - 2x^2 + 5) \, dx$
2. $\int \left(\frac{1}{x^3} - \sqrt{x}\right) dx$

Example

Integrate with respect to x :

1. $2\sin x - \cos x$
2. $-\dfrac{2}{x} + 3e^x$

EXERCISE 18

Integrate with respect to x:

a $3\sin x - 2$ b $4x - 2\cos x$ c $\sin x - 2\cos x + e^x$

d $x^2\sqrt{x} - 10\sin x$ e $\dfrac{x(x-1)}{3} + \cos x$ f $-\sin x + 2\sqrt{x}$

EXERCISE 19

Find:

a $\int \left(\sqrt{x} + \frac{1}{2}\cos x\right)\, dx$ b $\int \left(2e^t - 4\sin t\right)\, dt$ c $\int \left(3\cos t - \frac{1}{t}\right)\, dt$

EXERCISE 20

Find y if:

a $\dfrac{dy}{dx} = 6$ b $\dfrac{dy}{dx} = 4x^2$ c $\dfrac{dy}{dx} = 5\sqrt{x} - x^2$

d $\dfrac{dy}{dx} = \dfrac{1}{x^2}$ e $\dfrac{dy}{dx} = 2e^x - 5$ f $\dfrac{dy}{dx} = 4x^3 + 3x^2$

c) Particular value given

Example

Find $f(x)$ given that $f'(x) = x^3 - 2x^2 + 3$ and $f(0) = 2$.

Example

Find $f(x)$ given that $f'(x) = 2\sin x - \sqrt{x}$ and $f(0) = 4$.

Example

Find $f(x)$ given that $f''(x) = 12x^2 - 4$, $f'(0) = -1$, and $f(1) = 4$.

EXERCISE 21

Find $f(x)$ given that:

a $f'(x) = 2x - 1$ and $f(0) = 3$

b $f'(x) = 3x^2 + 2x$ and $f(2) = 5$

c $f'(x) = e^x + \frac{1}{\sqrt{x}}$ and $f(1) = 1$

d $f'(x) = x - \frac{2}{\sqrt{x}}$ and $f(1) = 2$

EXERCISE 22

Find $f(x)$ given that:

a $f''(x) = 2x + 1$, $f'(1) = 3$, and $f(2) = 7$

b $f''(x) = 15\sqrt{x} + \frac{3}{\sqrt{x}}$, $f'(1) = 12$, and $f(0) = 5$

c $f''(x) = \cos x$, $f'(\frac{\pi}{2}) = 0$, and $f(0) = 3$

d $f''(x) = 2x$ and the points $(1, 0)$ and $(0, 5)$ lie on the curve.

d) Integrating $f(ax+b)$

e) Substitution

IV- Definite integrals